Product and Quotient Rules Discovery

We have discussed the functions $y = x^2$ and $y = x^3$ before. We have already learned that the derivative of a sum (or difference) is the sum (or difference) of the derivatives.

Let's look at the derivative of a product $f(x) = g(x)h(x)$ and see if it works the same way. Here are the derivative charts for $y = x^2$ and $y = x^3$.

x	$f(x) = x^2$	$f'(x) = 2x$
-5	25	-10
-3	9	-6
-1	1	-2
0	0	0
2	4	4
4	16	8
6	36	12

Table 1

Note: There are several notations for representing the derivative (or slope) of a function. One of the most common is the "prime" notation that you see in the third column. In this notation, $f'(x)$ is read as "*f* prime of *x*" which means the derivative of f or the slope of f at point x.

x	$g(x) = x^3$	$g'(x) = 3x^2$
-5	-125	75
-3	-27	27
-1	-1	3
0	0	0
2	8	12
4	64	48
6	216	108

Table 2

Here is the table for $h(x) = x^2 \cdot x^3$.

Notice that in the second column, we multiplied the *y*-values for each function together. Based on what we did with sums and differences, multiply the derivative values together to complete the chart below:

x	$h(x)=x^2x^3$	$f'(x) \cdot g'(x)$
-5	-3125	
-3	-243	
-1	-1	
0	0	
2	32	
4	1024	
6	7776	

Table 3

Now, let's look at the problem in a different way. The product $x^2 \cdot x^3$ equals _____.
(Use the exponent laws to simplify the product.)

So, another way to write $h(x)$ is $h(x) =$ _____.

From this simplified form, write the formula for the derivative of the function, *h*, using the Power Rule that you learned earlier.

Complete the chart below using the formula you got in the previous step.

x	***h(x)=x²x³=x⁵***	***h '(x)***
-5	-3125	
-3	-243	
-1	-1	
0	0	
2	32	
4	1024	
6	7776	

Table 4

Do your charts match? (Of course not! Hopefully, you can see what the Product Rule isn't.)

Here are the derivative charts for $y = x^5$ and $y = x^3$. Fill in the derivative values from the previous chart (Table 4) that you made using the power rule.

x	***h(x) = x^5***	***h'(x)***
-5	-3125	
-3	-243	
-1	-1	
0	0	
2	32	
4	1024	
6	7776	

Table 5

x	***g(x) = x^3***	***g'(x)***
-5	-125	75
-3	-27	27
-1	-1	3
0	0	0
2	8	12
4	64	48
6	216	108

Table 6

Here is the table for $j(x) = \frac{x^5}{x^3}$.

Notice that in the second column, we divided the *y*-values for each function. Based on what we did with sums and differences, divide the derivative values to complete the chart below.

x	***j(x)=x^5/x^3***	***h'(x)/g'(x)***
-5	25	
-3	9	
-1	1	
0	undefined	
2	4	
4	16	
6	36	

Table 7

Let's examine the problem another way. The quotient $\frac{x^5}{x^3}$ equals _____. (Use the exponent laws to simplify the quotient.)

So, another way to write $j(x)$ is $j(x) =$ _____.

From this simplified form, write the formula for the derivative of the function, *h*, using the Power Rule that you learned earlier.

Complete the chart below using your formula for the derivative of the quotient.

x	***j(x)=x⁵/x³=x²***	***j'(x)***
-5	25	
-3	9	
-1	1	
0	0	
2	4	
4	16	
6	36	

Table 8

Do your charts match? (Of course not! Hopefully, you can see what the Quotient Rule isn't.)

Please note that you still don't know the product and quotient rules. But, you do know that to find the slope (or derivative) of a product of functions, you **don't** multiply the slopes. Likewise, you know that to find the slope (or derivative) of a quotient of functions, you **don't** divide the slopes.

In the next Activity you will learn how to find both the derivative of a product and the derivative of a quotient.